

Brutal Diagnosis - All Algorithms & Models

CRITICAL (7 items)

1. 18 models missing casts()

ChatMessage, CountyCultureSite, CountyFarm, CountyHealthFacility, CountyTourismAttraction, CountyTransport, CouponUsage, CourierShipment, CourierTrackingEvent, DisputeCase, EventBooking, ImageVariant, IncidentReport, JobApplication, NewsletterSubscriber, ReviewSeed, SafetyAlert, TimelineEvent

Impact: Data type mismatch, incorrect comparisons, silent failures

2. 10 models with `\$guarded = []`

ProductImage, ShoppingCart, ProductVariant, Supplier, ProductCategory, CartItem, TransactionLog, CountySubscriber, CountyBulkSlotAllocation, CountySubscriptionPlan

Impact: Mass-assignment vulnerability - anyone can mass-assign any column via request

3. CountyAdminController N+1 at line 666

SectorEntity::where('county_id',...)->get() inside loop without cache/pagination

Impact: Admin page loads ALL entities on every request

4. 10+ public routes without cache middleware

/counties/{slug}/sector/{sector}, /institutions/{institution}, /marketplace/{slug}, /experience/*, /flash-sales, /county/{slug}/subscriptions

Impact: Every page load hits DB + renders fresh

5. Route model binding without eager loading

OrderTrackingController::show, MarketplaceController::show, CountyController::show

Impact: N+1 queries on detail pages

6. VideoService zoom values hardcoded (1.15/1.0)

app/Services/VideoService.php:121-122

Impact: Zoom animation values not configurable

7. Missing foreign key indexes on permission tables

database/migrations/2026_07_22_110439_create_permission_tables.php

Impact: Slower JOIN queries on role/permission tables

HIGH (3 items)

8. CountyAdminController - no pagination on SectorEntity

CountyAdminController.php:666

Impact: Memory exhaustion on large counties

9. static `\$cache = []` in ThumbnailService

ThumbnailService.php:151

Impact: Memory leak in long-running processes

10. No CachePublicResponse on /experience/* routes

routes/web.php:88-95

Impact: POST routes acceptable but should be audited

LOW (3 items)

11. VideoService.zoomStart/End - 1.15/1.0 hardcoded (minor)

12. DisplayPriorityService comment still references +0.08 (already in config)

13. 10 `\$guarded = []` models - acceptable if inputs are validated

Previously Fixed (12 items)

- All 32 algorithm weights -> config/kicc.php
- Hardcoded county lists -> config
- resolveSlot() calls cached (6h TTL)
- in_array(\\$seen) $O(n^2)$ -> $O(1)$ with \\$seenMap
- Empty catch blocks -> logging (5 controllers)
- Sector tile posters pre-extracted in controller
- institution() product video query cached + simplified
- \\$countyHero dead code removed
- media-tile.js DOM destroy fix
- media-tile.blade.php template x-if removed
- Home page videos double handler removed
- County strip buffer clear removed